Chapter 14: G-Estimation of Structural Nested Models

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Chapters 12 and 13 estimated the average causal effect of smoking cessation on weight gain with IP weighting and standardization. This chapter describes the third **g-method**: **g-estimation**, applied to the same NHEFS data. Models whose parameters are estimated by g-estimation are called **structural nested models**.

This chapter is based on Hernán and Robins (2020, chap. 14, pp. 187-198).

Why g-methods? IP weighting, standardization, and g-estimation are called g-methods because they are designed for generalized treatment contrasts involving time-varying treatments. For time-fixed treatments they may be overkill, but the simple setting lets us focus on their main features before Part III.

Why only now? IP weighting and standardization appeared already in Part I (Chapter 2). G-estimation waits until Part II because it is easiest to describe with a structural model, even a saturated one. The three g-methods rest on different modeling assumptions.

1 14.1 The Causal Question Revisited (pp. 187-188)

Setting (as in Chapters 12-13): 1566 cigarette smokers aged 25-74 in NHEFS.

- Treatment A: quitting smoking ($A = 1$) vs. not ($A = 0$)
- Outcome Y: weight gain (kg)
- Covariates L: sex, age, race, education, intensity and duration of smoking, physical activity in daily life, recreational exercise, and weight
- Assumption: conditional exchangeability given L

Chapters 12-13 targeted the **population** average causal effect

$$\mathbb{E}\big[Y^{a=1,c=0}\big] - \mathbb{E}\big[Y^{a=0,c=0}\big]$$

the mean weight difference had everybody been treated and uncensored vs. untreated and uncensored.

The analysis is restricted to individuals with known sex, age, race, weight, height, education, alcohol use, and smoking intensity at baseline (1971-75) and follow-up (1982), who answered the baseline medical history questionnaire.

1.1 Effects Within Strata of L

This chapter instead targets the effect **within each stratum** of L:

$$\mathbb{E}\big[Y^{a=1,c=0} \mid L\big] - \mathbb{E}\big[Y^{a=0,c=0} \mid L\big]$$

- With extremely large data, one could estimate the effect separately in every stratum $L = l$ (Chapter 4), e.g., {non-quitter, female, white, age 26, college dropout, 15 cigarettes/day, 12 years of smoking, moderate exercise, very active, weight 112 kg}
- Alternatively, add all of L and all $A \times L$ product terms to the marginal structural model (Section 12.5): then $SW^A(L) = 1$ and the unweighted outcome regression, if correctly specified, adjusts for all confounding by L (Chapter 15)
- G-estimation of a **structural nested model** is a third route

For effects in a subset (e.g., individuals aged 45) one could also use marginal structural models with product terms (Chapter 12) or standardize to that subset only (Chapter 13).

2 14.2 Exchangeability Revisited (pp. 188-189)

Conditional exchangeability says that, within levels of L, the treated and untreated would have the same outcome distribution under the same treatment level:

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for } a = 0, 1$$

For $Y^{a=0}$: knowing $Y^{a=0}$ does not help distinguish quitters from non-quitters with the same L. For a dichotomous A this is equivalent to

$$\Pr[A = 1 \mid Y^{a=0}, L] = \Pr[A = 1 \mid L]$$

The book's margin note: if this paragraph feels repetitious given earlier discussions of exchangeability, "we could not be happier" (Hernán and Robins 2020, 188).

2.1 A Logistic Model Including the Counterfactual

$$\operatorname{logit} \Pr[A = 1 \mid Y^{a=0}, L] = \alpha_0 + \alpha_1 Y^{a=0} + \alpha_2 L$$

- $\alpha_2 L = \sum_{j=1}^{p} \alpha_{2j} L_j$ for $L = (L_1, \ldots, L_p)$
- Same as the model for the IP weight denominator in Chapter 12, plus $Y^{a=0}$ as a covariate

- We cannot fit it, since $Y^{a=0}$ is unknown for the treated

Question: If we could fit it, and exchangeability held and the model were correct, what would $\hat{\alpha}_1$ be?

Answer: its expected value is **zero**, because $Y^{a=0}$ does not predict A given L. This is half of g-estimation; the other half is the structural model.

3 14.3 Structural Nested Mean Models (pp. 189-191)

Target: $\mathbb{E}[Y^{a=1} \mid L] - \mathbb{E}[Y^{a=0} \mid L] = \mathbb{E}[Y^{a=1} - Y^{a=0} \mid L]$ (the difference of means equals the mean of differences). Ignore censoring for now.

No effect modification by L:

$$\mathbb{E}[Y^a - Y^{a=0} \mid L] = \beta_1 a$$

- No intercept β_0 and no main effect of L: these cancel in the difference

Effect modification by L (e.g., larger effect in heavy smokers):

$$\mathbb{E}[Y^a - Y^{a=0} \mid L] = \beta_1 a + \beta_2 aL$$

3.1 Conditioning on Treatment

Under $Y^a \perp\!\!\!\perp A \mid L$, the treated and untreated are the same type of people within levels of L, so the conditional effect is the same in both groups and we can write:

Definition 3.1 (Structural Nested Mean Model)**.**

$$\mathbb{E}[Y^a - Y^{a=0} \mid A = a, L] = \beta_1 a + \beta_2 aL$$

The parameters β_1 and β_2 (a vector), estimated by g-estimation, quantify the average causal effect of A on Y within levels of A and L.

Robins (1994) first described structural nested models. They are "nested" when treatment is time-varying (Part III).

3.2 Why "Semiparametric"?

- The parametric g-formula (Chapter 13) also models the outcome conditional on A and L
- But a structural nested model is **agnostic** about the intercept and the main effect of L: there is no β_0 and no $\beta_3 L$
- Leaving these unspecified means fewer assumptions and possibly more robustness to model misspecification than the parametric g-formula

i Fine Point 14.1: Marginal vs. Structural Nested Models

Consider the marginal structural model for a continuous covariate V (a component of L):

$$\mathbb{E}[Y^a \mid V] = \beta_0 + \beta_1 a + \beta_2 aV + \beta_3 V$$

- $\beta_1 + \beta_2 v = \mathbb{E}[Y^{a=1} - Y^{a=0} \mid V = v]$, the effect in individuals with $V = v$
- $\beta_0 + \beta_3 v = \mathbb{E}[Y^{a=0} \mid V = v]$

If only the effect is of interest, leave $\mathbb{E}[Y^{a=0} \mid V]$ unspecified:

$$\mathbb{E}[Y^a - Y^{a=0} \mid V] = \beta_1 a + \beta_2 aV$$

This **semiparametric marginal structural mean model** is more robust when V is

continuous or high-dimensional, because a misspecified model $\beta_0 + \beta_3 V$ can bias $(\hat{\beta}_1, \hat{\beta}_2)$ through correlated estimates.

- Conditional on a strict subset V of L, it equals a structural nested model for a "blip" of treatment given V; estimate it by g-estimation with V in place of L, weighting each individual by $SW^A(V) = f(A \mid V)/f(A \mid L)$
- Conditional on all of L ($SW^A = 1$), the "faux semiparametric MSM" $\mathbb{E}[Y^a - Y^{a=0} \mid L] = \beta_1 a + \beta_2 aL$ is, under conditional exchangeability, the structural nested mean model of this chapter

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3.3 Censoring

With censoring, the target is $\mathbb{E}[Y^{a=1,c=0} - Y^{a=0,c=0} \mid A, L]$, which requires adjusting for both confounding and selection bias.

- G-estimation adjusts for **confounding only**, not selection bias
- So first create a pseudo-population with nonstabilized IP weights for censoring, $W^C = 1/\Pr[C = 0 \mid L, A]$ (from Chapter 12), then g-estimate in that pseudo-population
- In the pseudo-population, the model is a structural model in the absence of censoring:

$$\mathbb{E}[Y^{a,c=0} - Y^{a=0,c=0} \mid A = a, L] = \beta_1 a + \beta_2 aL$$

The superscript $c = 0$ is omitted hereafter.

Technically, IP weighting for censoring is not necessary with g-estimation when treatment is time-fixed and does not affect any variable in L, and the outcome is measured at a single time: if $Y^a \perp\!\!\!\perp (A, C) \mid L$, g-estimation can be applied to the uncensored individuals without weights.

Continuous treatments: the model must specify the dose-response function, e.g., $\mathbb{E}[Y^a - Y^{a=0} \mid A = a, L] = \beta_1 a + \beta_2 a^2 + \beta_3 aL + \beta_4 a^2 L$, or a smooth (e.g., spline) function of A and L.

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i Technical Point 14.1: Multiplicative Structural Nested Mean Models

For positive outcomes, a multiplicative model is often preferred:

$$\log\left(\frac{\mathbb{E}[Y^a \mid A = a, L]}{\mathbb{E}[Y^{a=0} \mid A = a, L]}\right) = \beta_1 a + \beta_2 aL$$

It can be g-estimated with $H(\psi^\dagger) = Y \exp\{-\psi_1^\dagger a - \psi_2^\dagger aL\}$.

- For binary outcomes, this model was originally usable only for rare outcomes (to avoid predicted probabilities above 1); Richardson, Robins, and Wang (2017) removed that restriction, and the approach was extended to time-varying treatments with doubly robust g-estimation (Wang et al. 2022)
- The older alternative, a structural nested **logistic** model (logit contrast of Y^a vs. $Y^{a=0}$), is not collapsible (Fine Point 4.3) and does not generalize easily to time-varying treatments (Hernán and Robins 2020, 191)

4 14.4 Rank Preservation (pp. 191-193)

Rank all individuals by their observed Y: individual 23522 first (48.5 kg), 6928 second (47.5 kg), …, 23321 last (−41.3 kg). Now imagine ranking everyone by $Y^{a=1}$ and, separately, by $Y^{a=0}$.

Definition 4.1 (Rank Preservation)**. Rank preservation** holds if the ranking of individuals by $Y^{a=1}$ is identical to their ranking by $Y^{a=0}$.

- **Additive rank preservation**: the effect of A on Y is exactly the same, on the additive scale, for every individual
- **Conditional additive rank preservation**: the effect is exactly the same for all individuals with the same value of L

Example 4.1 (Rank-Preserving Structural Model)**.**

- If quitting increases everyone's weight by exactly 3 kg, additive rank preservation holds
- The sharp null hypothesis (Chapter 1) is a special case
- An (additive conditional) rank-preserving structural model:

$$Y_i^a - Y_i^{a=0} = \psi_1 a + \psi_2 a L_i \quad \text{for all individuals } i$$

so every individual with $L = l$ has $Y_i^{a=1} = Y_i^{a=0} + \psi_1 + \psi_2 l$

Figures 14.1-14.3: In Figure 14.1 (stratum $L = l$) every individual's outcome shifts by the same $\psi_1 + \psi_2 l$ from $Y^{a=0}$ to $Y^{a=1}$; Figure 14.2 shows another stratum $L = l'$ with a different outcome distribution and shift $\psi_1 + \psi_2 l'$. Figure 14.3 shows the realistic case: shifts vary across individuals with the same L, and ranks are not preserved (individual i is below j under $a = 0$ but not under $a = 1$).

4.1 Rank Preservation Is Implausible, and Not Needed

- Individual effects of quitting surely vary even within levels of L: some gain a lot, some little, some may lose weight
- **None of the methods in the book require rank preservation**: the MSM of Chapter 12 (estimate 3.5 kg, 95% CI 2.5 to 4.5) and the structural nested **mean** model of Section 14.3 are models for average effects
- The rank-preserving model is used only to **introduce** g-estimation, because g-estimation is easier to understand for it and the procedure is the same for rank-preserving and non-rank-preserving models

A structural nested mean model is well defined without rank preservation; for the setting of Figure 14.3, one can still model the average effect within strata of L. The rank-preserving model is a special structural mean model: the mean of the individual shifts within levels of L equals each individual shift.

5 14.5 G-Estimation (pp. 193-196)

Goal: estimate β_1 in $\mathbb{E}[Y^a - Y^{a=0} \mid A = a, L] = \beta_1 a$ (no effect modification by L).

Temporarily assume the rank-preserving model $Y_i^a - Y_i^{a=0} = \psi_1 a$ for all i, so $\psi_1 = \beta_1$. Rewrite it as

$$Y^{a=0} = Y^a - \psi_1 a$$

5.1 Linking the Model to the Observed Data

By consistency, $Y = Y^A$. Replacing a by each individual's observed A:

$$\begin{aligned} Y^{a=0} &= Y^A - \psi_1 A \quad (\text{set } a = A) \\ &= Y - \psi_1 A \quad (\text{consistency: } Y^A = Y) \end{aligned}$$

If we knew ψ_1 we could compute $Y^{a=0}$ for everyone. We don't.

5.2 The Guessing Game

A friend says ψ_1 is one of $\psi^\dagger \in \{-20, 0, 10\}$. For each candidate compute

$$H(\psi^\dagger) = Y - \psi^\dagger A$$

- $H(\psi^\dagger) = Y^{a=0}$ only when $\psi^\dagger = \psi_1$
- Under conditional exchangeability, $Y^{a=0}$ does not predict A given L (Section 14.2)
- So fit, for each candidate,

$$\text{logit}\Pr[A = 1 \mid H(\psi^\dagger), L] = \alpha_0 + \alpha_1 H(\psi^\dagger) + \alpha_2 L$$

- The candidate with $\alpha_1 = 0$ is $Y^{a=0}$; e.g., if $H(10)$ is unassociated with A given L, then $\hat{\psi}_1 = 10$

> **!** Important
>
> G-estimation does **not test** whether conditional exchangeability holds; it **assumes** it (Hernán and Robins 2020, 194).

Rosenbaum (1987) proposed a version of this procedure for non-time-varying treatments. Without a friend, we search over many values $\psi^\dagger$ (e.g., -20 to 20 in steps of 0.01); a finer search is closer to $\hat{\psi}_1$ but more computationally demanding.

5.3 NHEFS Results

Example 5.1 (G-Estimate of the Effect of Smoking Cessation (Program 14.2))**.**

- Computed 31 candidates $H(2.0), H(2.1), \ldots, H(5.0)$ and fit 31 logistic models (the Chapter 12 IP-weight denominator model plus $H(\psi^\dagger)$)
- $\hat{\alpha}_1$ was closest to zero for $H(3.4)$ and $H(3.5)$; a finer search found $\hat{\alpha}_1 \approx 0$ at $H(3.446)$
- **G-estimate**: $\hat{\psi}_1 = 3.4$ kg
- **95% CI**: 2.5 to 4.5 kg

5.4 Confidence Intervals by Test Inversion

- For each $\psi^\dagger$, test $\alpha_1 = 0$ (Wald test); at $\psi^\dagger = 3.446$ the P-value was 0.998
- The P-value exceeded 0.05 for $\psi^\dagger$ between about 2.5 and 4.5
- The 95% CI is the set of $\psi^\dagger$ with P-value > 0.05: **inversion** of the test

- Any valid test can be used; a score test avoids fitting multiple models and is essentially equivalent to the Wald test in large samples.
- With censoring, the logistic models are fit to uncensored individuals ($C = 0$), each weighted by the estimated IP weight for censoring (Technical Point 14.2).
- Because the test used a robust variance estimator (due to the IP weights), the 95% CI is conservative in large samples: it traps the true value at least 95% of the time. Bootstrapping the whole g-estimation procedure gives a non-conservative, possibly narrower, interval.

5.5 Back to Non-Rank-Preserving Models

The g-estimation algorithm for ψ_1 gives a **consistent estimate of** β_1 in the mean model whenever the mean model is correctly specified, **regardless of whether rank preservation holds**.

- It does not need $H(\beta_1) = Y^{a=0}$ for every individual
- It only needs $H(\beta_1)$ and $Y^{a=0}$ to have the same conditional mean given L

> **i** Fine Point 14.2: Sensitivity Analysis for Unmeasured Confounding
>
> G-estimation uses the fact that $\alpha_1 = 0$ under conditional exchangeability given L.
>
> - Suppose quitting is less likely when one's spouse smokes, and spouse's smoking is associated with determinants of weight gain not in L: unmeasured confounding, so

$\alpha_1 \neq 0$

- G-estimation does not actually require $\alpha_1 = 0$, only that the **magnitude of nonexchangeability** (the value of α_1) be known: if $\alpha_1 = 0.1$, search for the $\psi^\dagger$ at which $\hat{\alpha}_1 = 0.1$
- Repeating the analysis over a range of α_1 values and plotting the estimates gives a **sensitivity analysis** for unmeasured confounding
- A practical difficulty: how much confounding is $\alpha_1 = 0.1$? See Robins, Rotnitzky, and Scharfstein (1999) for technical details

6 14.6 Structural Nested Models with Two or More Parameters (pp. 196-197)

Without the product term $\beta_2 aL$, the model assumes no effect modification by L.

- If some components V of L modify the effect and the term $\beta_2 aV$ is omitted, the structural nested model is **misspecified** and generally biased
- Contrast: the saturated MSM $\mathbb{E}[Y^a] = \beta_0 + \beta_1 a$ is not misspecified when it omits $\beta_2 aV$ and $\beta_3 V$, because it targets the population effect
- Structural nested models, by definition, target effects **within levels of** L

Null preservation: when the null of no average causal effect is true, marginal structural models (Chapter 12) and structural nested models are never misspecified. The parametric g-formula preserves the null for time-fixed treatments but not in the time-varying setting (Part III).

6.1 Two Parameters

Suppose the effect depends on baseline smoking intensity V:

$$\mathbb{E}[Y^a - Y^{a=0} \mid A = a, L] = \beta_1 a + \beta_2 aV$$

- Candidate: $\beta^\dagger = (\beta_1^\dagger, \beta_2^\dagger)$ and $H(\beta^\dagger) = Y - \beta_1^\dagger A - \beta_2^\dagger AV$
- Fit the IP-weighted logistic model

$$\text{logit}\Pr[A = 1 \mid H(\beta^\dagger), L] = \alpha_0 + \alpha_1 H(\beta^\dagger) + \alpha_2 H(\beta^\dagger)V + \alpha_3 L$$

- Find $(\beta_1^\dagger, \beta_2^\dagger)$ making **both** α_1 and α_2 zero: a two-dimensional search

6.2 No Search Needed in General

- For **linear** structural nested mean models the estimator has **closed form** (Technical Point 14.2)
- For nonlinear models, g-estimation solves an **estimating equation**, so derivative-based methods (e.g., Newton-Raphson) apply
- Some structural nested survival models need a search, because their estimating equation is not differentiable (Chapter 17)

Example 6.1 (Two-Parameter Model in NHEFS (Program 14.3))**.** G-estimates: $\hat{\beta}_1 = 2.86$ and $\hat{\beta}_2 = 0.03$. Confidence intervals are most easily obtained by bootstrapping.

6.3 Effect Modification by All of L

For dichotomous A, the unsaturated model

$$\mathbb{E}[Y^a - Y^{a=0} \mid A = a, L] = \beta_1 a + a \sum_{j=1}^{p} \beta_{2j} L_j$$

has $p+1$ parameters. The average causal effect in the study population is then estimated by averaging the stratum-specific effect over the n individuals:

$$\frac{1}{n}\sum_{i=1}^{n}\left(\beta_1+\sum_{j=1}^{p}\beta_{2j}L_{ij}\right)=\frac{1}{n}\sum_{i=1}^{n}\beta_1+\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{p}\beta_{2j}L_{ij}$$
$$=\beta_1+\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{p}\beta_{2j}L_{ij}$$

The book writes the left-hand side of this model as $\mathbb{E}[Y^a - Y^{a=0}]$; the conditioning on $A=a, L$ is implied by the right-hand side depending on L (Hernán and Robins 2020, 197).

Next: after the three g-methods, Chapter 15 reviews arguably the most common approaches to confounding adjustment: outcome regression and propensity scores.

Technical Point 14.2: G-Estimation of Structural Nested Mean Models

General model: $\mathbb{E}[Y - Y^{a=0} \mid A, L] = A\gamma(L;\beta)$, with $\gamma(L;\beta^\dagger)$ known and $\gamma(L;0)=0$.
Estimating equation. Choose $\beta^\dagger$ minimizing the association between $H(\beta^\dagger) = Y - A\gamma(L;\beta^\dagger)$ and A given L. Basing this on a score test means solving

$$\sum_{i=1}^{n} I[C_i=0]\, W_i^C\, H_i(\beta^\dagger)(A_i - \mathbb{E}[A \mid L_i])q(L_i) = 0$$

where $q(L_i)$ is a user-chosen vector function of the same dimension as β, and W_i^C and $\mathbb{E}[A \mid L_i] = \Pr[A=1 \mid L_i]$ are replaced by estimates. The choice of q affects efficiency (CI length), not consistency; Robins (1994) derived the choice that minimizes CI length.
Closed form. If $\gamma(L;\beta^\dagger) = \beta^{\dagger T} d(L)$ for a known $d(L)$, choose $q(L) = d(L)$ and write $R_i = I[C_i=0]W_i^C(A_i - \mathbb{E}[A \mid L_i])$. Then

$$0 = \sum_i R_i(Y_i - A_i\, d(L_i)^T\beta)d(L_i) \qquad \text{(substitute } H_i;\ \beta^T d = d^T\beta)$$
$$\sum_i R_i A_i\, d(L_i)\, d(L_i)^T\beta = \sum_i R_i Y_i\, d(L_i) \qquad \text{(move the } \beta \text{ term to the left)}$$
$$\hat\beta = \left(\sum_i R_i A_i\, d(L_i)\, d(L_i)^T\right)^{-1}\sum_i R_i Y_i\, d(L_i) \qquad \text{(solve for } \beta)$$

Mean vs. distributional independence. Structural nested mean models only impose $\mathbb{E}[H(\beta_1) \mid A, L] = \mathbb{E}[H(\beta_1) \mid L]$, so nonlinear functions of $H_i(\beta^\dagger)$ (e.g., its cube) cannot be used in the estimating equation; they can be for **structural nested distribution models**, which impose $H(\beta_1) \perp\!\!\!\perp A \mid L$.
Double robustness. The estimator above is consistent only if the models for $\mathbb{E}[A \mid L]$ and $\Pr[C=1 \mid A, L]$ are both correct. Replacing $H(\beta^\dagger)$ by $H(\beta^\dagger) - \mathbb{E}[H(\beta^\dagger) \mid L]$, with the latter estimated by an unweighted linear model among the uncensored, gives an estimator that is consistent if either (i) the model for $\mathbb{E}[H(\beta^\dagger) \mid L]$ or (ii) both the treatment and censoring models are correct: a **doubly robust** estimator (Hernán and Robins 2020, 198).

7 Summary

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1. **Structural nested mean models** $\mathbb{E}[Y^a - Y^{a=0} \mid A=a, L] = \beta_1 a + \beta_2 aL$ model the effect within levels of L and leave $\mathbb{E}[Y^{a=0} \mid L]$ unspecified (semiparametric).
2. **Exchangeability** can be written as: $Y^{a=0}$ does not predict A given L ($\alpha_1 = 0$).
3. **G-estimation** finds the parameter value whose candidate counterfactual $H(\psi^\dagger)$ is unassociated with A given L; NHEFS: 3.4 kg (95% CI 2.5, 4.5).

4. **Rank preservation** is implausible and **not required**: it only simplifies the explanation.
5. **Censoring** is handled by IP weights; g-estimation itself adjusts only for confounding.
6. Models with **effect modification** need product terms; linear models have a closed-form g-estimator, which can be made **doubly robust**.

8 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.